Phishing_Random_Forests

April 17, 2024

[1]:
```
# The Supervised Machine Learning Process (Classification algorithms)
##########################################  Step 1 Data Wrangling &␣
↪prerocessing #####################
# Step 1 Part 0 Installing machine learning libraries and data wrangling␣
↪libraries = Done
# Step 1 Part 1 Importing libraries = Done
# Step 1 part 2 Importing dataset = Done (imported into and sql database and␣
↪then export into csv)
# Step 1 part 3 Taking care of missing data and duplicates = Done (no missing␣
↪data or duplicates)
# Step 1 part 4 Encoding categorical data = Done, no need for missing but the␣
↪dataset is checked!
# Step 1 part 5 Feature selection (creating a matrix of features, and the␣
↪dependant variable we are predicting = Done, machine learning process␣
↪expects these two entities)
# Step 1 part 6 Splitting dataset into training and test
# Step 1 part 7 Feature scaling

########################################## Step 3 Random Forests␣
↪############################

# Step 3 Part 0
# Step 3 Part 1
# Step 3 part 2
######################################### THE END ############################


################ Resources and Datasets #######################

###########https://data.mendeley.com/datasets/6tm2d6sz7p/1 ###### Dataset.csv␣
↪#### Published: 7 June 2023 (0=legitimate,1=phishing)
#1. df_Dataset = pd.read_csv('datasets/Dataset.csv')

#2. df_phishing_csv = pd.read_csv('datasets/phishing_dataset.csv',␣
↪on_bad_lines='skip')# https://gregavrbancic.github.io/Phishing-Dataset/#␣
↪2020 dataset
```

```
################# Machine learning resoruces
# https://www.youtube.com/watch?v=i_LwzRVP7bg&t=2687s
```

[ ]:

```

```

[2]:

```
# Step 1 Part 0 Installing machine learning libraries and data wrangling␣
↪libraries = Done

import sys
!{sys.executable} -m pip install pandas
!{sys.executable} -m pip install matplotlib
!{sys.executable} -m pip install numpy
!{sys.executable} -m pip install scikit-learn
!{sys.executable}pip install xgboost
!{sys.executable} pip install CMake
```

```
Requirement already satisfied: pandas in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (1.5.3)
Requirement already satisfied: python-dateutil>=2.8.1 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from pandas)
(2023.3.post1)
Requirement already satisfied: numpy>=1.21.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from pandas) (1.24.3)
Requirement already satisfied: six>=1.5 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from python-
dateutil>=2.8.1->pandas) (1.16.0)
Requirement already satisfied: matplotlib in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (3.7.2)
Requirement already satisfied: contourpy>=1.0.1 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (1.0.5)
Requirement already satisfied: cycler>=0.10 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: numpy>=1.20 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (1.24.3)
Requirement already satisfied: packaging>=20.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (23.1)
Requirement already satisfied: pillow>=6.2.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (10.0.1)
Requirement already satisfied: pyparsing<3.1,>=2.3.1 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from matplotlib) (2.8.2)
```

```
Requirement already satisfied: six>=1.5 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from python-
dateutil>=2.7->matplotlib) (1.16.0)
Requirement already satisfied: numpy in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (1.24.3)
Requirement already satisfied: scikit-learn in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (1.1.3)
Requirement already satisfied: numpy>=1.17.3 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from scikit-learn)
(1.24.3)
Requirement already satisfied: scipy>=1.3.2 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from scikit-learn)
(1.10.0)
Requirement already satisfied: joblib>=1.0.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from scikit-learn) (1.2.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in
/opt/homebrew/anaconda3/lib/python3.11/site-packages (from scikit-learn) (2.2.0)
zsh:1: no such file or directory: /opt/homebrew/anaconda3/bin/pythonpip
/opt/homebrew/anaconda3/bin/python: can't open file
'/Users/perceval/data_science/machine_learning/msc_project/pip': [Errno 2] No
such file or directory
```

[3]:
```python
# Step 1 Part 1 Importing libraries = Done

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
import joblib
from joblib import dump
```

[4]:
```python
# Step 1 part 2 Importing dataset, creating a matrix of features, and the␣
↪dependant variable we are predicting = Done, machine learning process␣
↪expects these two entities
#df_phishing_csv = pd.read_csv('datasets/phishing_dataset.csv',␣
↪on_bad_lines='skip')# https://gregavrbancic.github.io/Phishing-Dataset/#␣
↪2020 dataset

df_phishing_csv = pd.read_csv('datasets/Dataset.csv', on_bad_lines='skip') #␣
↪https://data.mendeley.com/datasets/6tm2d6sz7p/1 #2023 dataset with␣
↪0=legitimate, 1 = phishing
```

[5]:
```python
has_nan = df_phishing_csv.isnull().values.any()
has_nan
```

[5]: False

[6]:
```python
cleaned_df = df_phishing_csv.dropna(how='any')
cleaned_df
```

[6]:
```
        Type  url_length  number_of_dots_in_url  \
0          0          37                      2
1          1          70                      5
2          0          42                      2
3          0          46                      2
4          0          51                      3
…        …          …                      …
247945     0          42                      1
247946     0          42                      2
247947     1          33                      2
247948     1          83                      1
247949     0          34                      3

        having_repeated_digits_in_url  number_of_digits_in_url  \
0                                   0                        0
1                                   0                        0
2                                   0                        6
3                                   0                        0
4                                   0                        0
…                                 …                        …
247945                              0                        0
247946                              0                        0
247947                              0                        0
247948                              1                       19
247949                              0                        0

        number_of_special_char_in_url  number_of_hyphens_in_url  \
0                                   8                         0
1                                  12                         0
2                                   8                         0
3                                   7                         0
4                                   9                         0
…                                 …                        …
247945                              6                         0
247946                              8                         0
247947                              8                         0
247948                              9                         0
247949                              7                         0

        number_of_underline_in_url  number_of_slash_in_url  \
0                                0                       5
1                                0                       6
2                                0                       3
3                                0                       4
```

```
4                                   0                        5
…                                 …                        …
247945                              0                        4
247946                              0                        5
247947                              0                        5
247948                              0                        7
247949                              0                        3

        number_of_questionmark_in_url  …  having_digits_in_subdomain  \
0                                   0  …                            0
1                                   0  …                            0
2                                   1  …                            0
3                                   0  …                            0
4                                   0  …                            0
…                                 …  …                          …
247945                              0  …                            0
247946                              0  …                            0
247947                              0  …                            0
247948                              0  …                            0
247949                              0  …                            0

        number_of_digits_in_subdomain  having_repeated_digits_in_subdomain  \
0                                   0                                     1
1                                   0                                     1
2                                   0                                     1
3                                   0                                     1
4                                   0                                     1
…                                 …                                     …
247945                              0                                     1
247946                              0                                     1
247947                              0                                     1
247948                              0                                     1
247949                              0                                     1

        having_path  path_length  having_query  having_fragment  \
0                 0            3             0                0
1                 0            4             0                0
2                 0            1             1                0
3                 0            2             0                0
4                 0            3             0                0
…               …            …             …                …
247945            0            2             0                0
247946            0            3             0                0
247947            0            3             0                0
247948            0            5             0                0
247949            0            1             0                0
```

```
        having_anchor  entropy_of_url  entropy_of_domain
0                   0        4.010412           2.751629
1                   0        4.089470           3.532573
2                   0        4.386016           3.344698
3                   0        4.221947           3.189898
4                   0        4.103538           2.952820
…                 …              …                  …
247945              0        3.758289           3.323231
247946              0        3.937093           3.026987
247947              0        3.813207           3.327820
247948              0        4.540173           3.375000
247949              0        4.035274           3.182006

[247950 rows x 42 columns]
```

[7]:
```
# Step 1 part 2 preview dataset
cleaned_df
```

[7]:
```
        Type  url_length  number_of_dots_in_url  \
0          0          37                      2
1          1          70                      5
2          0          42                      2
3          0          46                      2
4          0          51                      3
…       …          …                      …
247945     0          42                      1
247946     0          42                      2
247947     1          33                      2
247948     1          83                      1
247949     0          34                      3

        having_repeated_digits_in_url  number_of_digits_in_url  \
0                                   0                        0
1                                   0                        0
2                                   0                        6
3                                   0                        0
4                                   0                        0
…                                 …                        …
247945                              0                        0
247946                              0                        0
247947                              0                        0
247948                              1                       19
247949                              0                        0

        number_of_special_char_in_url  number_of_hyphens_in_url  \
0                                   8                         0
1                                  12                         0
```

```
2                                   8                          0
3                                   7                          0
4                                   9                          0
…                                 …                        …
247945                              6                          0
247946                              8                          0
247947                              8                          0
247948                              9                          0
247949                              7                          0

        number_of_underline_in_url  number_of_slash_in_url  \
0                                0                        5
1                                0                        6
2                                0                        3
3                                0                        4
4                                0                        5
…                              …                        …
247945                           0                        4
247946                           0                        5
247947                           0                        5
247948                           0                        7
247949                           0                        3

        number_of_questionmark_in_url  …  having_digits_in_subdomain  \
0                                   0  …                            0
1                                   0  …                            0
2                                   1  …                            0
3                                   0  …                            0
4                                   0  …                            0
…                                 …  …                          …
247945                              0  …                            0
247946                              0  …                            0
247947                              0  …                            0
247948                              0  …                            0
247949                              0  …                            0

        number_of_digits_in_subdomain  having_repeated_digits_in_subdomain  \
0                                   0                                     1
1                                   0                                     1
2                                   0                                     1
3                                   0                                     1
4                                   0                                     1
…                                 …                                     …
247945                              0                                     1
247946                              0                                     1
247947                              0                                     1
247948                              0                                     1
```

```
247949                                0                                     1

        having_path  path_length  having_query  having_fragment  \
0                 0            3             0                0
1                 0            4             0                0
2                 0            1             1                0
3                 0            2             0                0
4                 0            3             0                0
…               …            …             …                …
247945            0            2             0                0
247946            0            3             0                0
247947            0            3             0                0
247948            0            5             0                0
247949            0            1             0                0

        having_anchor  entropy_of_url  entropy_of_domain
0                   0        4.010412           2.751629
1                   0        4.089470           3.532573
2                   0        4.386016           3.344698
3                   0        4.221947           3.189898
4                   0        4.103538           2.952820
…                 …              …                  …
247945              0        3.758289           3.323231
247946              0        3.937093           3.026987
247947              0        3.813207           3.327820
247948              0        4.540173           3.375000
247949              0        4.035274           3.182006

[247950 rows x 42 columns]
```

[8]:
```python
# Step 1 part 2 #describe dataset
cleaned_df.describe()
```

[8]:
```
                Type     url_length  number_of_dots_in_url  \
count  247950.000000  247950.000000          247950.000000
mean        0.481585      58.358197               2.894075
std         0.499662      29.764620               1.690970
min         0.000000      12.000000               0.000000
25%         0.000000      39.000000               2.000000
50%         0.000000      49.000000               3.000000
75%         1.000000      69.000000               3.000000
max         1.000000     191.000000              28.000000

       having_repeated_digits_in_url  number_of_digits_in_url  \
count                  247950.000000            247950.000000
mean                       0.162351                 4.366070
std                        0.368773                 9.579032
```

```
min                          0.000000                 0.000000
25%                          0.000000                 0.000000
50%                          0.000000                 0.000000
75%                          0.000000                 4.000000
max                          1.000000               144.000000

       number_of_special_char_in_url  number_of_hyphens_in_url  \
count                  247950.000000             247950.000000
mean                       10.303218                  0.600867
std                         4.718381                  1.235599
min                         4.000000                  0.000000
25%                         7.000000                  0.000000
50%                         9.000000                  0.000000
75%                        11.000000                  1.000000
max                        75.000000                 23.000000

       number_of_underline_in_url  number_of_slash_in_url  \
count               247950.000000           247950.000000
mean                     0.204049                4.784533
std                      0.653502                1.812574
min                      0.000000                2.000000
25%                      0.000000                4.000000
50%                      0.000000                4.000000
75%                      0.000000                5.000000
max                     47.000000               30.000000

       number_of_questionmark_in_url  …  having_digits_in_subdomain  \
count                  247950.000000  …               247950.000000
mean                       0.159911  …                    0.064529
std                        0.449506  …                    0.245694
min                        0.000000  …                    0.000000
25%                        0.000000  …                    0.000000
50%                        0.000000  …                    0.000000
75%                        0.000000  …                    0.000000
max                       17.000000  …                    1.000000

       number_of_digits_in_subdomain  having_repeated_digits_in_subdomain  \
count                  247950.000000                             247950.0
mean                       0.224013                                  1.0
std                        1.276447                                  0.0
min                        0.000000                                  1.0
25%                        0.000000                                  1.0
50%                        0.000000                                  1.0
75%                        0.000000                                  1.0
max                       44.000000                                  1.0

       having_path    path_length   having_query  having_fragment  \
```

```
count     247950.0  247950.000000  247950.000000    247950.000000
mean           0.0       2.606977       0.126501         0.002303
std            0.0       1.636938       0.332414         0.047933
min            0.0       0.000000       0.000000         0.000000
25%            0.0       2.000000       0.000000         0.000000
50%            0.0       2.000000       0.000000         0.000000
75%            0.0       3.000000       0.000000         0.000000
max            0.0      28.000000       1.000000         1.000000

       having_anchor  entropy_of_url  entropy_of_domain
count  247950.000000   247950.000000      247950.000000
mean        0.000944        4.257222           3.389839
std         0.030706        0.328098           0.376361
min         0.000000        2.648939           1.386274
25%         0.000000        4.038910           3.180833
50%         0.000000        4.208298           3.392747
75%         0.000000        4.428036           3.606937
max         1.000000        5.866327           4.957323

[8 rows x 42 columns]
```

[9]:
```
# Step 1 part 2 #number of records and features (247950 records with 42␣
↪features)
cleaned_df.shape
```

[9]:
```
(247950, 42)
```

[10]:
```
# Step 1 part 2 #information about columns and data types
cleaned_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 247950 entries, 0 to 247949
Data columns (total 42 columns):
 #   Column                                      Non-Null Count   Dtype
---  ------                                      --------------   -----
 0   Type                                        247950 non-null  int64
 1   url_length                                  247950 non-null  int64
 2   number_of_dots_in_url                       247950 non-null  int64
 3   having_repeated_digits_in_url               247950 non-null  int64
 4   number_of_digits_in_url                     247950 non-null  int64
 5   number_of_special_char_in_url               247950 non-null  int64
 6   number_of_hyphens_in_url                    247950 non-null  int64
 7   number_of_underline_in_url                  247950 non-null  int64
 8   number_of_slash_in_url                      247950 non-null  int64
 9   number_of_questionmark_in_url               247950 non-null  int64
 10  number_of_equal_in_url                      247950 non-null  int64
 11  number_of_at_in_url                         247950 non-null  int64
 12  number_of_dollar_in_url                     247950 non-null  int64
```

```
 13  number_of_exclamation_in_url                247950 non-null  int64
 14  number_of_hashtag_in_url                    247950 non-null  int64
 15  number_of_percent_in_url                    247950 non-null  int64
 16  domain_length                               247950 non-null  int64
 17  number_of_dots_in_domain                    247950 non-null  int64
 18  number_of_hyphens_in_domain                 247950 non-null  int64
 19  having_special_characters_in_domain         247950 non-null  int64
 20  number_of_special_characters_in_domain      247950 non-null  int64
 21  having_digits_in_domain                     247950 non-null  int64
 22  number_of_digits_in_domain                  247950 non-null  int64
 23  having_repeated_digits_in_domain            247950 non-null  int64
 24  number_of_subdomains                        247950 non-null  int64
 25  having_dot_in_subdomain                     247950 non-null  int64
 26  having_hyphen_in_subdomain                  247950 non-null  int64
 27  average_subdomain_length                    247950 non-null  float64
 28  average_number_of_dots_in_subdomain         247950 non-null  int64
 29  average_number_of_hyphens_in_subdomain      247950 non-null  int64
 30  having_special_characters_in_subdomain      247950 non-null  int64
 31  number_of_special_characters_in_subdomain   247950 non-null  int64
 32  having_digits_in_subdomain                  247950 non-null  int64
 33  number_of_digits_in_subdomain               247950 non-null  int64
 34  having_repeated_digits_in_subdomain         247950 non-null  int64
 35  having_path                                 247950 non-null  int64
 36  path_length                                 247950 non-null  int64
 37  having_query                                247950 non-null  int64
 38  having_fragment                             247950 non-null  int64
 39  having_anchor                               247950 non-null  int64
 40  entropy_of_url                              247950 non-null  float64
 41  entropy_of_domain                           247950 non-null  float64
dtypes: float64(3), int64(39)
memory usage: 79.5 MB
```

[11]:
```python
# Step 1 part 3 Taking care of missing data
# Check for missing values
print(cleaned_df.isnull().sum())
```

```
Type                                        0
url_length                                  0
number_of_dots_in_url                       0
having_repeated_digits_in_url               0
number_of_digits_in_url                     0
number_of_special_char_in_url               0
number_of_hyphens_in_url                    0
number_of_underline_in_url                  0
number_of_slash_in_url                      0
number_of_questionmark_in_url               0
number_of_equal_in_url                      0
number_of_at_in_url                         0
```

```
number_of_dollar_in_url                      0
number_of_exclamation_in_url                 0
number_of_hashtag_in_url                     0
number_of_percent_in_url                     0
domain_length                                0
number_of_dots_in_domain                     0
number_of_hyphens_in_domain                  0
having_special_characters_in_domain          0
number_of_special_characters_in_domain       0
having_digits_in_domain                      0
number_of_digits_in_domain                   0
having_repeated_digits_in_domain             0
number_of_subdomains                         0
having_dot_in_subdomain                      0
having_hyphen_in_subdomain                   0
average_subdomain_length                     0
average_number_of_dots_in_subdomain          0
average_number_of_hyphens_in_subdomain       0
having_special_characters_in_subdomain       0
number_of_special_characters_in_subdomain    0
having_digits_in_subdomain                   0
number_of_digits_in_subdomain                0
having_repeated_digits_in_subdomain          0
having_path                                  0
path_length                                  0
having_query                                 0
having_fragment                              0
having_anchor                                0
entropy_of_url                               0
entropy_of_domain                            0
dtype: int64
```

[12]:
```python
# Step 1 part 3a Taking care of missing data
print(cleaned_df.isnull().sum())
```

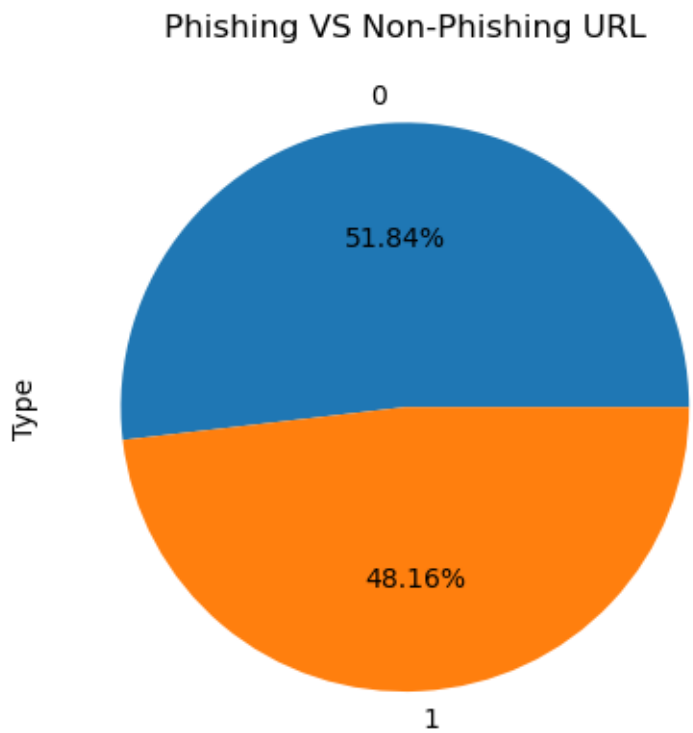
```
Type                                         0
url_length                                   0
number_of_dots_in_url                        0
having_repeated_digits_in_url                0
number_of_digits_in_url                      0
number_of_special_char_in_url                0
number_of_hyphens_in_url                     0
number_of_underline_in_url                   0
number_of_slash_in_url                       0
number_of_questionmark_in_url                0
number_of_equal_in_url                       0
number_of_at_in_url                          0
number_of_dollar_in_url                      0
```

```
number_of_exclamation_in_url                 0
number_of_hashtag_in_url                     0
number_of_percent_in_url                     0
domain_length                                0
number_of_dots_in_domain                     0
number_of_hyphens_in_domain                  0
having_special_characters_in_domain          0
number_of_special_characters_in_domain       0
having_digits_in_domain                      0
number_of_digits_in_domain                   0
having_repeated_digits_in_domain             0
number_of_subdomains                         0
having_dot_in_subdomain                      0
having_hyphen_in_subdomain                   0
average_subdomain_length                     0
average_number_of_dots_in_subdomain          0
average_number_of_hyphens_in_subdomain       0
having_special_characters_in_subdomain       0
number_of_special_characters_in_subdomain    0
having_digits_in_subdomain                   0
number_of_digits_in_subdomain                0
having_repeated_digits_in_subdomain          0
having_path                                  0
path_length                                  0
having_query                                 0
having_fragment                              0
having_anchor                                0
entropy_of_url                               0
entropy_of_domain                            0
dtype: int64
```

[13]:
```python
# Step 1 part 3b Taking care of missing data
#checking for unique records on each column
print(cleaned_df['Type'].nunique())
```

```
2
```

[14]:
```python
#Step 1 part 3c Distribution of phishing links to non phishing links
# Count in a pie chart
cleaned_df['Type'].value_counts().plot(kind='pie', autopct='%1.2f%%')
plt.title("Phishing VS Non-Phishing URL")
plt.show()

# print(cleaned_df['Type'].nunique())
```



[15]:
```
# Step 1 part 3d #distribution of categorical data in column result,

print(cleaned_df['Type'].value_counts())
```

```
0    128541
1    119409
Name: Type, dtype: int64
```

[16]:
```
# Step 1 part 3e Taking care of missing data
print(cleaned_df.isnull().sum())
```

```
Type                                        0
url_length                                  0
number_of_dots_in_url                       0
having_repeated_digits_in_url               0
number_of_digits_in_url                     0
number_of_special_char_in_url               0
number_of_hyphens_in_url                    0
number_of_underline_in_url                  0
number_of_slash_in_url                      0
number_of_questionmark_in_url               0
number_of_equal_in_url                      0
number_of_at_in_url                         0
```

```
number_of_dollar_in_url                      0
number_of_exclamation_in_url                 0
number_of_hashtag_in_url                     0
number_of_percent_in_url                     0
domain_length                                0
number_of_dots_in_domain                     0
number_of_hyphens_in_domain                  0
having_special_characters_in_domain          0
number_of_special_characters_in_domain       0
having_digits_in_domain                      0
number_of_digits_in_domain                   0
having_repeated_digits_in_domain             0
number_of_subdomains                         0
having_dot_in_subdomain                      0
having_hyphen_in_subdomain                   0
average_subdomain_length                     0
average_number_of_dots_in_subdomain          0
average_number_of_hyphens_in_subdomain       0
having_special_characters_in_subdomain       0
number_of_special_characters_in_subdomain    0
having_digits_in_subdomain                   0
number_of_digits_in_subdomain                0
having_repeated_digits_in_subdomain          0
having_path                                  0
path_length                                  0
having_query                                 0
having_fragment                              0
having_anchor                                0
entropy_of_url                               0
entropy_of_domain                            0
dtype: int64
```

[17]:
```
# Step 1 part 3f Taking care of missing data
#checking for unique records on each column
print(cleaned_df['Type'].nunique())
```

```
2
```

[18]:
```
# Step 1 part 4 #distribution of categorical data in column result,
```

[19]:
```
# Step 1 part 4 Encoding categorical data
#distribution of categorical data in column Type
#Column Type has already 1 & 0 therefore there is no need to encode the data␣
↪into categorical values
print(cleaned_df['Type'].value_counts())
```

```
0    128541
1    119409
Name: Type, dtype: int64
```

[20]:
```
# Step 1 part 5 Feature selection (creating a matrix of features from the␣
↪dataset)

X = cleaned_df.iloc[:, 1:].values #get all valriable (features) except the␣
↪first column
y = cleaned_df.iloc[:, 0].values #get all pridictive values from the first␣
↪column
```

[21]:
```
print(X)
```

```
[[37.          2.          0.          …  0.          4.01041207
   2.75162917]
 [70.          5.          0.          …  0.          4.08946998
   3.53257326]
 [42.          2.          0.          …  0.          4.38601593
   3.34469838]
 …
 [33.          2.          0.          …  0.          3.81320653
   3.32781953]
 [83.          1.          1.          …  0.          4.54017264
   3.375     ]
 [34.          3.          0.          …  0.          4.0352735
   3.18200582]]
```

[22]:
```
print(y)
```

```
[0 1 0 … 1 1 0]
```

[ ]:
```

```

[23]:
```
# Step 1 part 6 (Option 1) Splitting dataset into training and test

# valid dataset i.e used to ensure that the model can  handle unseen data after␣
↪training
# training dataset i.e used to train the model
# test dataset i.e use to evaluate the model
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2,␣
↪random_state = 0)
```

[24]:
```
# Step 1 part 7 feature scaling to improve training perfomance
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

[25]:
```
# Import the model we are using
#Fitting Decision Tree classifier to the training set
```

```
from sklearn.ensemble import RandomForestClassifier
classifier= RandomForestClassifier(n_estimators= 10, criterion="entropy")
classifier.fit(X_train, y_train)
```

[25]:
```
RandomForestClassifier(criterion='entropy', n_estimators=10)
```

[26]:
```
#Predicting the Test set result
y_pred = classifier.predict(X_test)
```

[27]:
```
y_pred
```

[27]:
```
array([1, 0, 0, …, 0, 1, 1])
```

[28]:
```
#Creating the Confusion matrix
from sklearn.metrics import confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
```

[29]:
```
cm
```

[29]:
```
array([[25080,   667],
       [ 1243, 22600]])
```

[ ]:
```

```

[30]:
```
# from the above matrix
# incorect predictions => 1261 + 661 = 1 922
# Correct predictions => 25086 + 22582 = 47 668
# True Positive 22582
# True Negative 25086
```

[31]:
```
# import the metrics class
from sklearn.metrics import classification_report, accuracy_score,␣
↪confusion_matrix, precision_score, recall_score, f1_score,␣
↪roc_auc_score,matthews_corrcoef,r2_score

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.95      0.97      0.96     25747
           1       0.97      0.95      0.96     23843

    accuracy                           0.96     49590
   macro avg       0.96      0.96      0.96     49590
weighted avg       0.96      0.96      0.96     49590
```

[32]:
```
print("Accuracy")
print(accuracy_score(y_test, y_pred))
```

```
Accuracy
0.961484170195604
```

[33]:
```
print("Precision Score")
print(precision_score(y_test, y_pred))
```

```
Precision Score
0.9713327889285254
```

[34]:
```
print("Recall score")
print(recall_score(y_test, y_pred))
```

```
Recall score
0.9478672985781991
```

[35]:
```
print("F1-score")
print(f1_score(y_test, y_pred))
```

```
F1-score
0.9594565909573338
```

[36]:
```
print("ROC Score")
print(roc_auc_score(y_test, y_pred))
```

```
ROC Score
0.9609806838950731
```

[37]:
```
print("MCC")
print(matthews_corrcoef(y_test, y_pred))
```

```
MCC
0.9230359214441343
```

[ ]:
```

```

[38]:
```
########################################### important features␣
↪#####################
feature_importances = classifier.feature_importances_
```

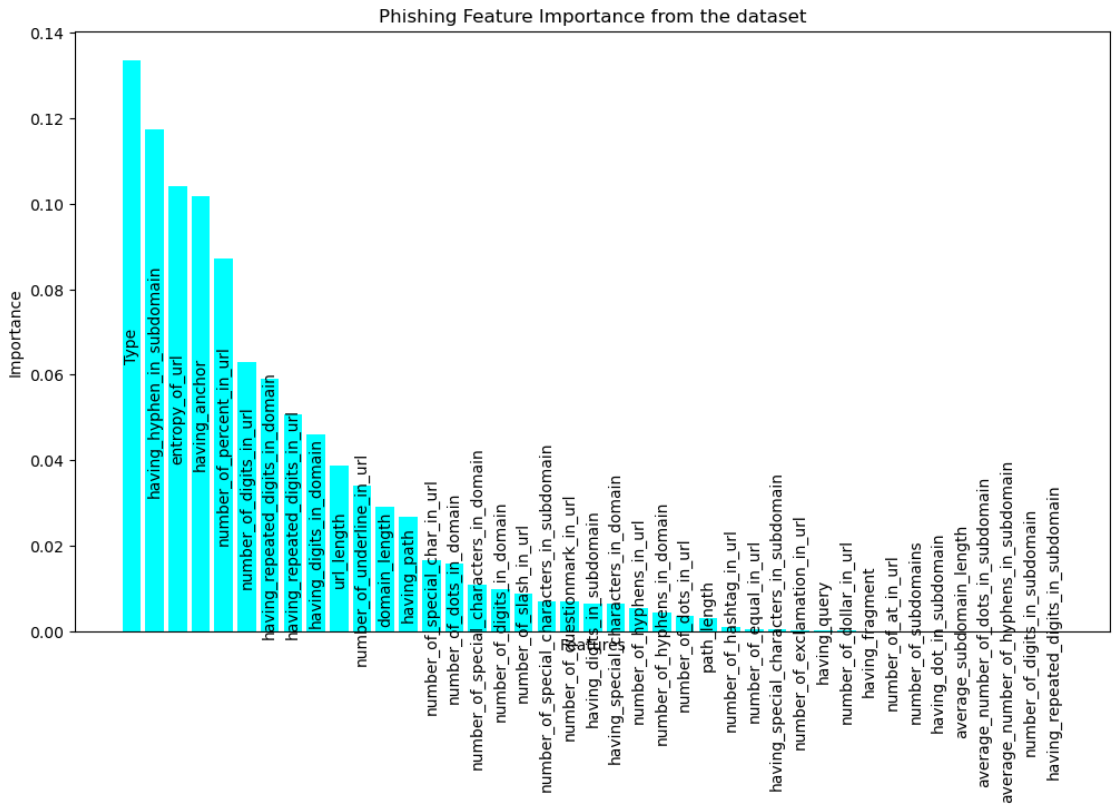
[39]:
```
feature_importance_pairs = zip(cleaned_df.columns, feature_importances)
```

[40]:
```
sorted_features = sorted(feature_importance_pairs, key=lambda x: x[1],␣
↪reverse=True)
```

[41]:
```
# Prin
```

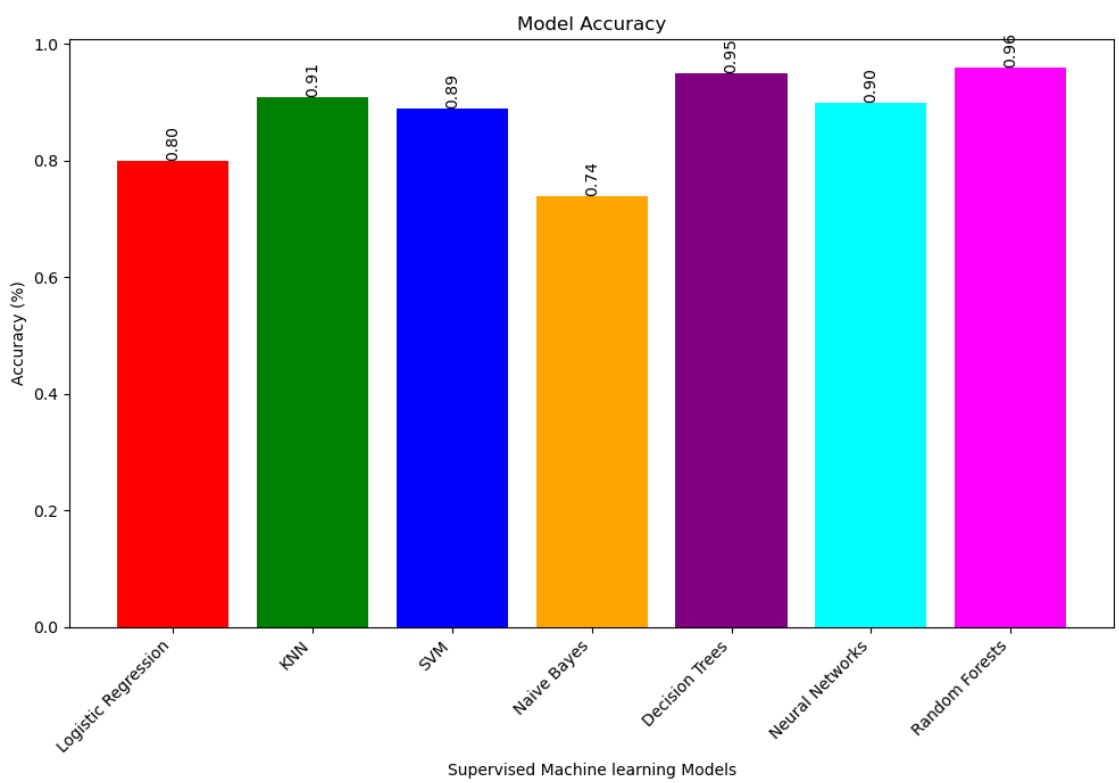
[42]:
```
labels, values = zip(*sorted_features)
```

[43]:
```
plt.figure(figsize=(12, 7))
bars = plt.bar(labels, values, color='cyan')
plt.xlabel('Features')
plt.ylabel('Importance')
plt.title('Phishing Feature Importance from the dataset')

# Displaying feature titles vertically
for bar, label in zip(bars, labels):
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval/2, label, ha='center',␣
 ↪va='center', color='black', rotation=90)

# Removing the labels at the bottom
plt.xticks([])

plt.show()
```

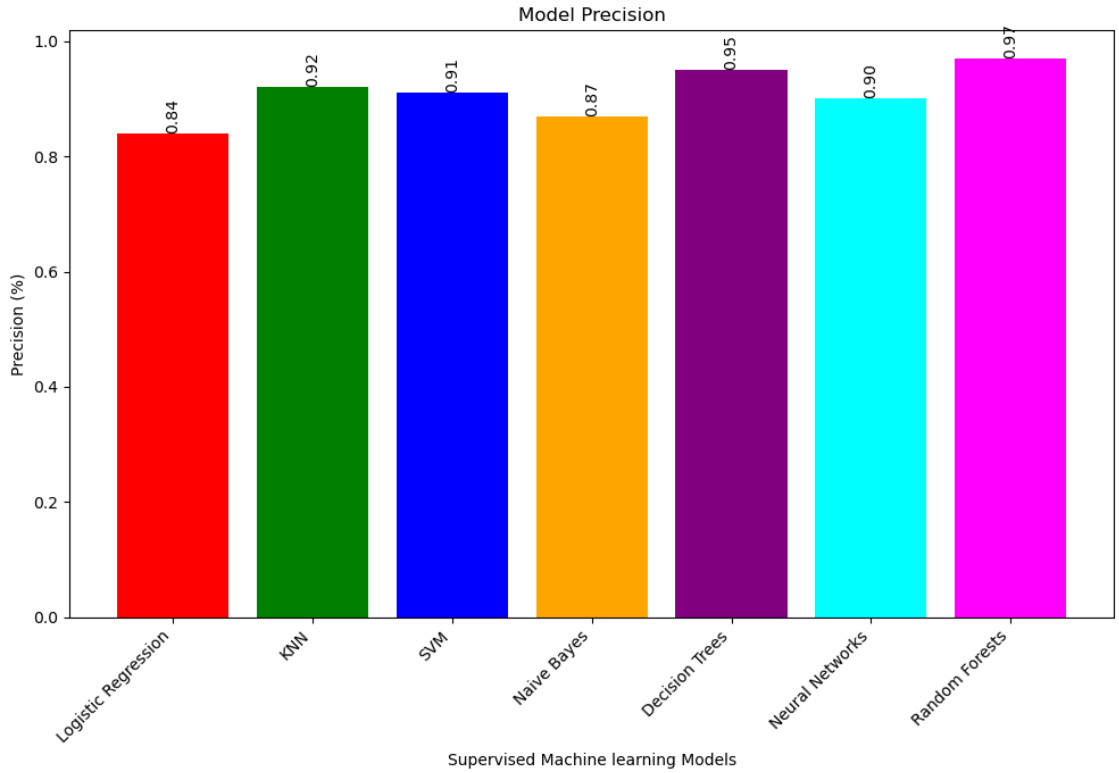


[44]:
```
# Defining the accuracy data
accuracy_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
```

```
    'Accuracy': [0.80, 0.91, 0.89, 0.74, 0.95, 0.90, 0.96]
}

# Creating the DataFrame again
accuracy_df = pd.DataFrame(accuracy_data)

# Plotting a bar graph with different colors. Extending the color list to match␣
↪the number of categories.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(accuracy_df['Models'], accuracy_df['Accuracy'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
↪verticalalignment='bottom', ha='center', rotation=90)

plt.title('Model Accuracy')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('Accuracy (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```

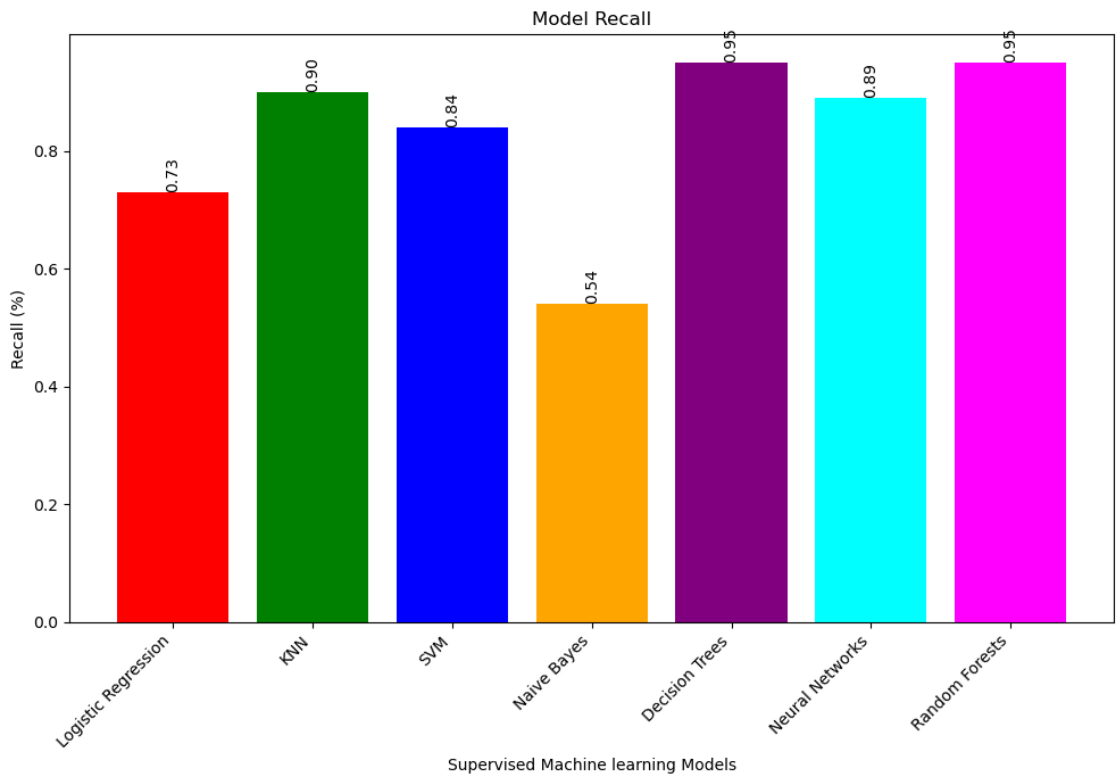


[45]:
```
# Ranking by precision

# Defining the accuracy data
precision_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
    'Precision': [0.84, 0.92, 0.91, 0.87, 0.95, 0.90, 0.97]
}

# Creating the DataFrame again
precision_df = pd.DataFrame(precision_data)

# Plotting a bar graph with different colors. Extending the color list to match␣
 ↪the number of categories.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(precision_df['Models'], precision_df['Precision'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
 ↪verticalalignment='bottom', ha='center', rotation=90)

plt.title('Model Precision')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('Precision (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```



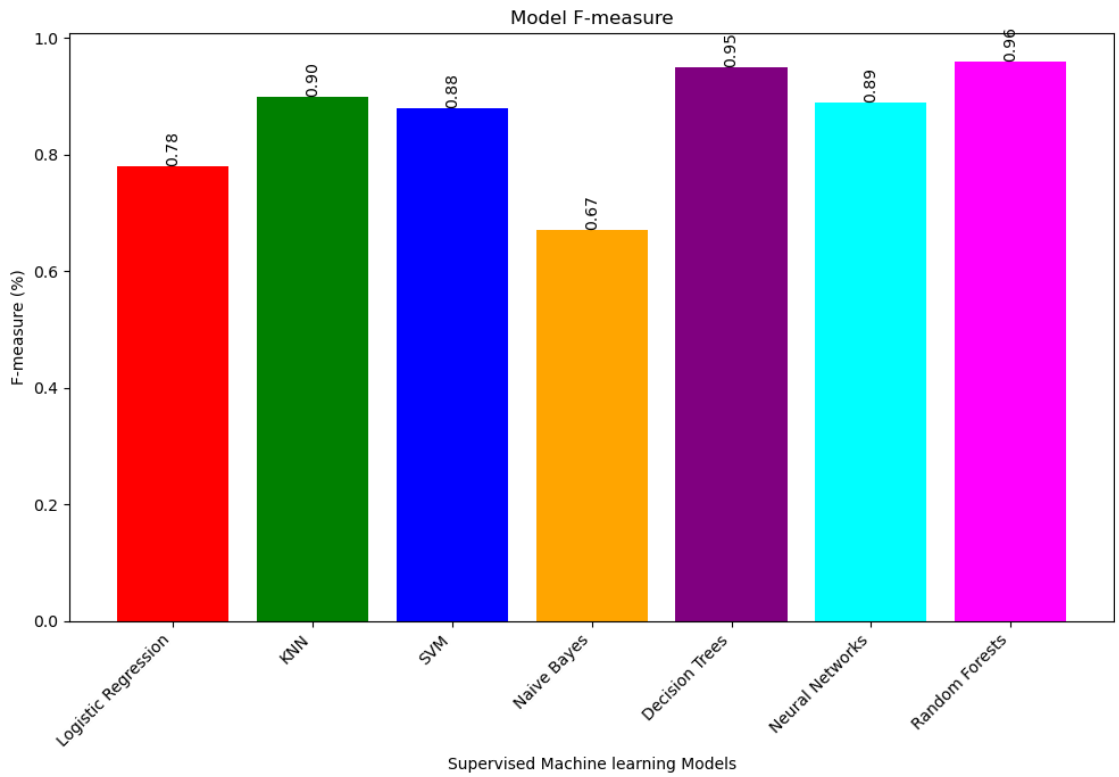
[46]:
```
# Ranking by recall

recall_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
    'Recall': [0.73, 0.90, 0.84, 0.54, 0.95, 0.89, 0.95]
}

# Creating the DataFrame again
recall_df = pd.DataFrame(recall_data)

# Plotting a bar graph with different colors. Extending the color list to match␣
 ↪the number of categories.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(recall_df['Models'], recall_df['Recall'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
 ↪verticalalignment='bottom', ha='center', rotation=90)
```

```
plt.title('Model Recall')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('Recall (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```



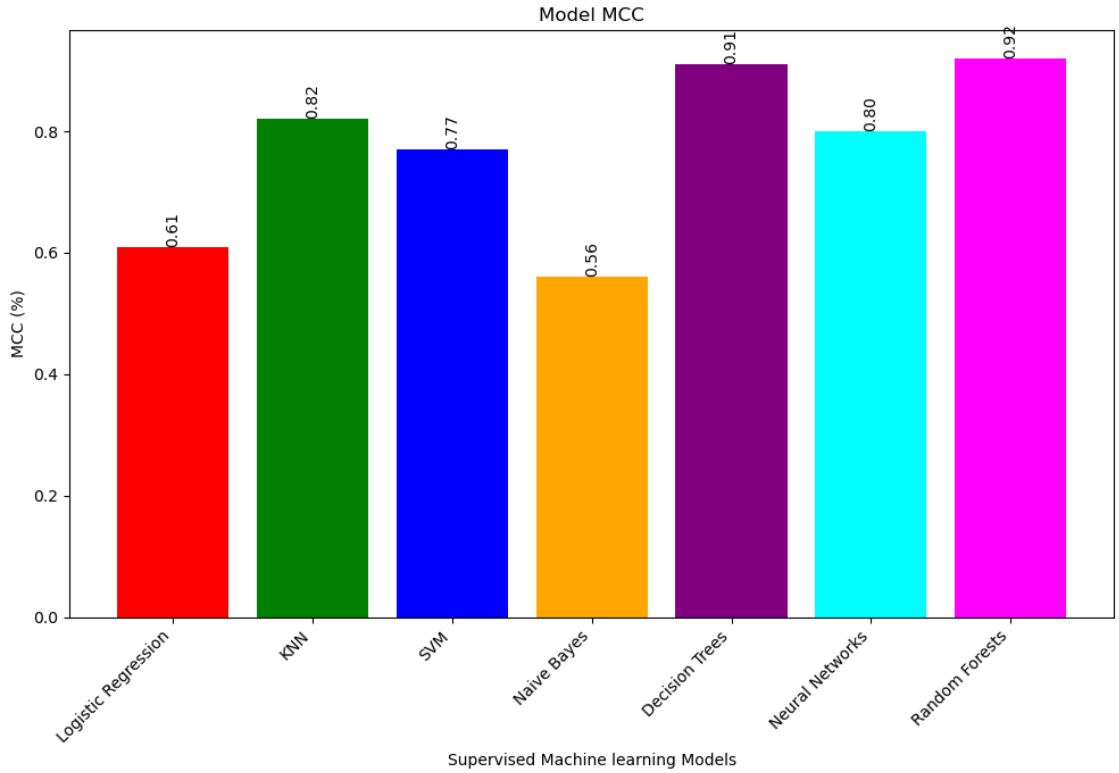
[47]:
```
# Ranking by F-measure


# Defining the fmeasure
fmeasure_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
    'F-measure': [0.78, 0.90, 0.88, 0.67, 0.95, 0.89, 0.96]
}

# Creating the DataFrame again
fmeasure_df = pd.DataFrame(fmeasure_data)
```

```
# Plotting a bar graph with different colors. Extending the color list to match␣
↪the number of categories.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(fmeasure_df['Models'], fmeasure_df['F-measure'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
↪verticalalignment='bottom', ha='center', rotation=90)

plt.title('Model F-measure')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('F-measure (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```



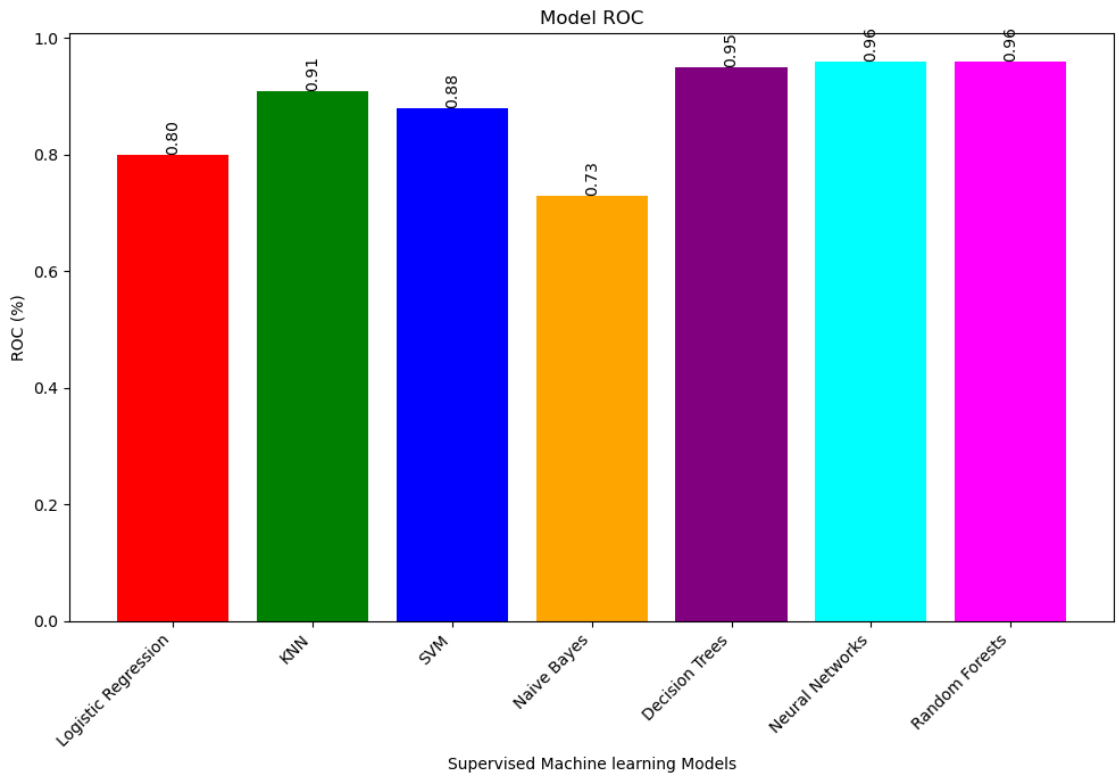
[48]:
```
# Ranking by Mathews Correlation Coefficient

# Defining the MCC data
```

```
mcc_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
    'MCC': [0.61, 0.82, 0.77, 0.56, 0.91, 0.80, 0.92]
}

# Creating the DataFrame again
mcc_df = pd.DataFrame(mcc_data)

# Plotting a bar graph with different colors. Extending the color list to match␣
 ↪the number of categories.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(mcc_df['Models'], mcc_df['MCC'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
 ↪verticalalignment='bottom', ha='center', rotation=90)

plt.title('Model MCC')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('MCC (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```



[49]:
```
# Ranking by Receiver Operating Characteristic (ROC) Area

# Ranking by ROC

# Defining the MCC data
roc_data = {
    'Models': ['Logistic Regression','KNN','SVM','Naive Bayes','Decision␣
 ↪Trees', 'Neural Networks', 'Random Forests'],
    'ROC': [0.80, 0.91, 0.88, 0.73, 0.95, 0.96, 0.96]
}

# Creating the ROC DataFrame
roc_df = pd.DataFrame(roc_data)

# Plotting a bar graph with different colors.
colors = ['red', 'green', 'blue', 'orange', 'purple', 'cyan', 'magenta']
plt.figure(figsize=(10, 7))
bars = plt.bar(roc_df['Models'], roc_df['ROC'], color=colors)

# Adding accuracy scores to the top of the bars
for bar in bars:
    yval = bar.get_height()
```

```
    plt.text(bar.get_x() + bar.get_width()/2.0, yval, f'{yval:.2f}',␣
 ↪verticalalignment='bottom', ha='center', rotation=90)

plt.title('Model ROC')
plt.xlabel('Supervised Machine learning Models')
plt.ylabel('ROC (%)')
plt.xticks(rotation=45, ha='right')  # Rotate model names for better readability
plt.tight_layout()  # Adjust layout to make room for rotated labels
plt.show()
```



[50]:
```
joblib.dump(classifier, '/Users/perceval/data_science/machine_learning/
 ↪msc_project/phishing_detect/phishing_random_forest_model.pkl')
```

[50]: ['/Users/perceval/data_science/machine_learning/msc_project/phishing_detect/phishing_random_forest_model.pkl']

[51]:
```
dump(classifier, '/Users/perceval/data_science/machine_learning/msc_project/
 ↪phishing_detect/phishing_detector.joblib')
```

[51]: ['/Users/perceval/data_science/machine_learning/msc_project/phishing_detect/phishing_detector.joblib']

[52]:
```
#building the model again with a few features for the application
cleaned_df
```

[52]:
```
        Type  url_length  number_of_dots_in_url  \
0          0          37                      2
1          1          70                      5
2          0          42                      2
3          0          46                      2
4          0          51                      3
…        …          …                      …
247945     0          42                      1
247946     0          42                      2
247947     1          33                      2
247948     1          83                      1
247949     0          34                      3

        having_repeated_digits_in_url  number_of_digits_in_url  \
0                                   0                        0
1                                   0                        0
2                                   0                        6
3                                   0                        0
4                                   0                        0
…                                 …                        …
247945                              0                        0
247946                              0                        0
247947                              0                        0
247948                              1                       19
247949                              0                        0

        number_of_special_char_in_url  number_of_hyphens_in_url  \
0                                   8                         0
1                                  12                         0
2                                   8                         0
3                                   7                         0
4                                   9                         0
…                                 …                        …
247945                              6                         0
247946                              8                         0
247947                              8                         0
247948                              9                         0
247949                              7                         0

        number_of_underline_in_url  number_of_slash_in_url  \
0                                0                       5
1                                0                       6
2                                0                       3
3                                0                       4
```

```
4                                   0                        5
…                                 …                        …
247945                              0                        4
247946                              0                        5
247947                              0                        5
247948                              0                        7
247949                              0                        3

        number_of_questionmark_in_url  …  having_digits_in_subdomain  \
0                                   0  …                            0
1                                   0  …                            0
2                                   1  …                            0
3                                   0  …                            0
4                                   0  …                            0
…                                 …  …                          …
247945                              0  …                            0
247946                              0  …                            0
247947                              0  …                            0
247948                              0  …                            0
247949                              0  …                            0

        number_of_digits_in_subdomain  having_repeated_digits_in_subdomain  \
0                                   0                                     1
1                                   0                                     1
2                                   0                                     1
3                                   0                                     1
4                                   0                                     1
…                                 …                                     …
247945                              0                                     1
247946                              0                                     1
247947                              0                                     1
247948                              0                                     1
247949                              0                                     1

        having_path  path_length  having_query  having_fragment  \
0                 0            3             0                0
1                 0            4             0                0
2                 0            1             1                0
3                 0            2             0                0
4                 0            3             0                0
…               …            …             …                …
247945            0            2             0                0
247946            0            3             0                0
247947            0            3             0                0
247948            0            5             0                0
247949            0            1             0                0
```

```
        having_anchor  entropy_of_url  entropy_of_domain
0                   0        4.010412           2.751629
1                   0        4.089470           3.532573
2                   0        4.386016           3.344698
3                   0        4.221947           3.189898
4                   0        4.103538           2.952820
…                 …              …                  …
247945              0        3.758289           3.323231
247946              0        3.937093           3.026987
247947              0        3.813207           3.327820
247948              0        4.540173           3.375000
247949              0        4.035274           3.182006

[247950 rows x 42 columns]
```

[53]:
```
df_django =␣
↪cleaned_df[['Type','having_anchor','entropy_of_url','having_hyphen_in_subdomain','number_of
```

[54]:
```
df_django
```

[54]:
```
        Type  having_anchor  entropy_of_url  having_hyphen_in_subdomain  \
0          0              0        4.010412                           0
1          1              0        4.089470                           0
2          0              0        4.386016                           0
3          0              0        4.221947                           0
4          0              0        4.103538                           0
…       …             …              …                           …
247945     0              0        3.758289                           0
247946     0              0        3.937093                           0
247947     1              0        3.813207                           0
247948     1              0        4.540173                           0
247949     0              0        4.035274                           0

        number_of_percent_in_url  having_repeated_digits_in_domain  \
0                              0                                  0
1                              0                                  0
2                              0                                  0
3                              0                                  0
4                              0                                  0
…                            …                                  …
247945                         0                                  0
247946                         0                                  0
247947                         0                                  0
247948                         0                                  0
247949                         0                                  0

        number_of_digits_in_url  having_repeated_digits_in_url  domain_length  \
```

```
0                                0                                0              12
1                                0                                0              26
2                                6                                0              21
3                                0                                0              15
4                                0                                0              16
…                              …                              …              …
247945                           0                                0              15
247946                           0                                0              13
247947                           0                                0              16
247948                          19                                1              16
247949                           0                                0              14

        having_digits_in_domain  url_length  number_of_underline_in_url  \
0                              0          37                           0
1                              0          70                           0
2                              1          42                           0
3                              0          46                           0
4                              0          51                           0
…                            …           …                           …
247945                         0          42                           0
247946                         0          42                           0
247947                         0          33                           0
247948                         1          83                           0
247949                         0          34                           0

        having_path  number_of_dots_in_domain
0                 0                         2
1                 0                         5
2                 0                         2
3                 0                         2
4                 0                         3
…               …                         …
247945            0                         1
247946            0                         2
247947            0                         2
247948            0                         1
247949            0                         3

[247950 rows x 14 columns]
```

[55]:
```
# Step 1 again feature selection

XX = df_django.iloc[:, 1:].values #get all valriable (features) except the␣
↪first column
yy = df_django.iloc[:, 0].values #get all pridictive values from the first␣
↪column
```

[56]:
```
#split dataset
```

[57]:
```
# Step 2 split dataset for the second model

from sklearn.model_selection import train_test_split
XX_train, XX_test, yy_train, yy_test = train_test_split(XX, yy, test_size = 0.
↪2, random_state = 0)
```

[58]:
```
# Step 3 standard scaller
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
XX_train = sc.fit_transform(XX_train)
XX_test = sc.transform(XX_test)
```

[59]:
```
print(XX_test)
```

```
[[-0.03176925  1.6723372   0.         … -0.31178178  0.
   0.65847362]
 [-0.03176925 -0.14361757  0.         … -0.31178178  0.
  -0.52960895]
 [-0.03176925  0.01123972  0.         … -0.31178178  0.
  -0.52960895]
 …
 [-0.03176925 -0.50964065  0.         … -0.31178178  0.
  -0.52960895]
 [-0.03176925  1.1550665   0.         … -0.31178178  0.
   0.06443234]
 [-0.03176925  0.11879561  0.         … -0.31178178  0.
  -0.52960895]]
```

[60]:
```
print(XX_train)
```

```
[[-0.03176925 -1.01029264  0.         … -0.31178178  0.
   0.06443234]
 [-0.03176925 -0.22664633  0.         … -0.31178178  0.
  -0.52960895]
 [-0.03176925  0.16695165  0.         … -0.31178178  0.
   0.65847362]
 …
 [-0.03176925 -1.04452362  0.         … -0.31178178  0.
   0.65847362]
 [-0.03176925 -0.54574376  0.         … -0.31178178  0.
   0.06443234]
 [-0.03176925  1.23229164  0.         … -0.31178178  0.
  -0.52960895]]
```

[61]:
```
# Step 4 load model
#Fitting Decision Tree classifier to the training set
```

```python
from sklearn.ensemble import RandomForestClassifier
classifier2= RandomForestClassifier(n_estimators= 10, criterion="entropy")
classifier2.fit(XX_train, yy_train)
```

[61]: RandomForestClassifier(criterion='entropy', n_estimators=10)

[62]:
```python
# Step 5 second prediction
y_pred2 = classifier2.predict(XX_test)
```

[63]:
```python
# import the metrics class
from sklearn.metrics import classification_report, accuracy_score,␣
↪confusion_matrix, precision_score, recall_score, f1_score,␣
↪roc_auc_score,matthews_corrcoef,r2_score

print(classification_report(yy_test, y_pred2))
```

```
              precision    recall  f1-score   support

           0       0.90      0.93      0.92     25747
           1       0.92      0.89      0.91     23843

    accuracy                           0.91     49590
   macro avg       0.91      0.91      0.91     49590
weighted avg       0.91      0.91      0.91     49590
```

[64]:
```python
joblib.dump(classifier2, '/Users/perceval/data_science/machine_learning/
↪msc_project/phishing_detect/phishing_random_forest_model_2.pkl')
```

[64]: ['/Users/perceval/data_science/machine_learning/msc_project/phishing_detect/phishing_random_forest_model_2.pkl']

[65]:
```python
dump(classifier2, '/Users/perceval/data_science/machine_learning/msc_project/
↪phishing_detect/phishing_detector_2.joblib')
```

[65]: ['/Users/perceval/data_science/machine_learning/msc_project/phishing_detect/phishing_detector_2.joblib']